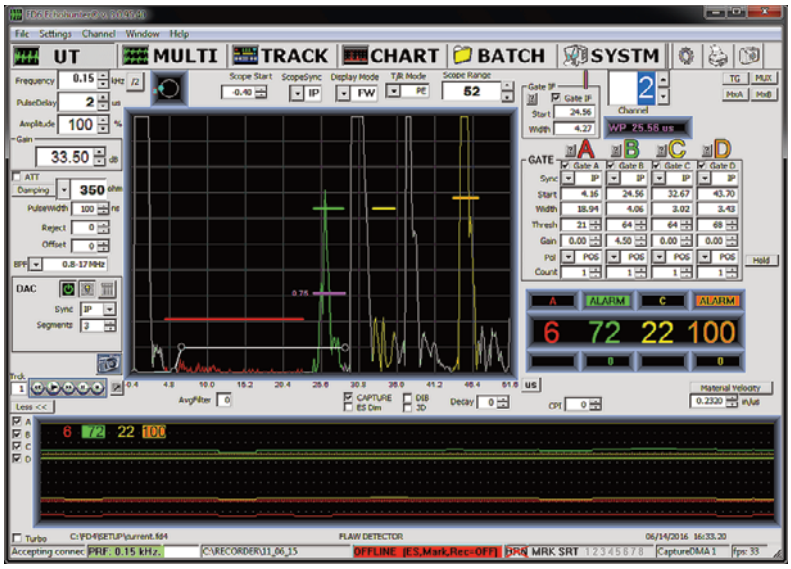
MAC®
Magnetic Analysis Corp.

Echomac® II

Ultrasonic Instrument for Flaw Detection, Thickness, and Dimensional Measurement in Tube & Bar



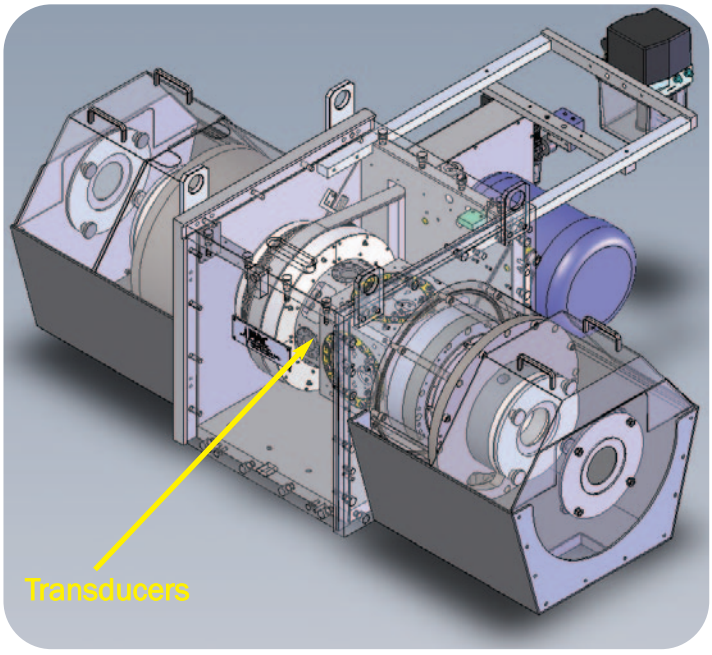
Features of Echomac Electronics



- ☑ Includes up to 16 independent test channels in a single computer.
- ☑ Increased Gain Range with fine resolution and improved linearity.
- ☑ User configurable criteria for flaw, lamination and thickness gauging, independently for each channel.
- ☑ Adjustable pulse firing sequence to avoid crosstalk in multi-channel applications.
- ☑ Four independent flaw gates for each channel with improved resolution.
- ☑ 15 segments distance amplitude correction (DAC).
- ☑ Improved DAC interface and resolution.
- ☑ Very high resolution thickness measurement for each channel.
- ☑ Programmable for Pulse Echo or Through Transmission.

Echomac II Applications

- ☑ Inspect carbon, duplex, or stainless steels, aluminum, titanium, copper and other metals and alloys.
- ☑ Detect flaws and measure dimensions and wall thickness.
- ☑ Test tube & bar for internal defects and inclusions.
- ☑ Inspect for tube ovality and eccentricity.
- ☑ Inspect strip before welding.
- ☑ Upgrade and/or replace older ultrasonic testers & systems.
- ☑ Use with rotary, spin-the-tube, squirter, bubbler installations.



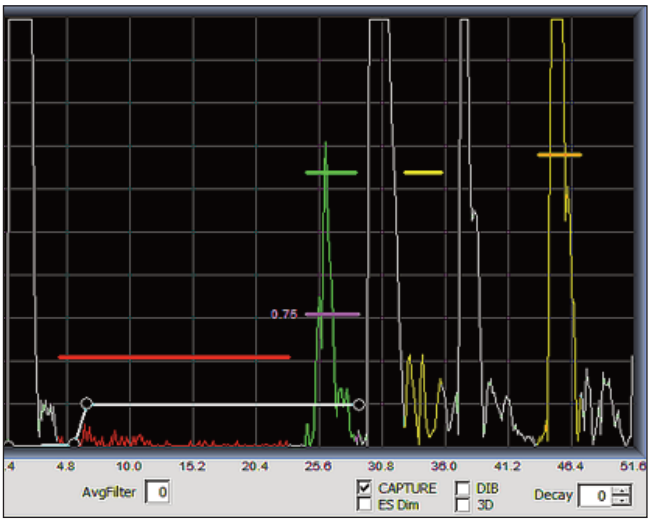
Echomac® Rotaries rotate up to 32 transducers around the tube or bar, as it is moved through the test. Water is continuously circulated in the transducer housing to maintain the couplant for the UT signals.

Magnetic Analysis Corporation ~ 103 Fairview Park Drive, Elmsford, New York 10523-1544 USA ~ Tel +1-914-530-2000 ~ info@mac-ndt.com
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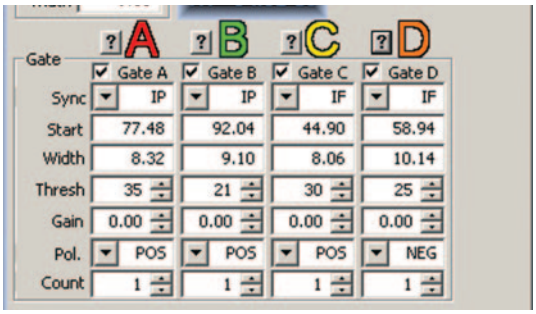
Inspection Features

Superior Performance

- ❒ Vivid, real time, flicker free full color display of test signals, thresholds and settings.
- ❒ High Signal-to-Noise ratio.
- ❒ 16 Independent Channels in one Instrument.
- ❒ 16 step damping adjustment for precise resolution.
- ❒ Wide choice of band pass filters.
- ❒ Negative square wave pulse echo or pitch-catch through transmission optimize transducer efficiency.
- ❒ Meets API & ASTM standards.
- ❒ Excellent repeatability of test results.



A-Scan display in the UT Screen shows the setup for Channel 1 with a gate interface and 4 gate thresholds.



Gate Dialog Box in the UT Screen

Versatile, Intuitive Operation

- ❒ Set up and control all key test parameters on one screen with a click of the mouse or keyboard.
- ❒ Move thresholds by selecting and dragging on screen.
- ❒ Adjust parameters for several channels at once with the “Global” key. Or easily copy a group of test parameters from one channel to another.
- ❒ Follow up test results and supervise operators remotely.
- ❒ Versatile, robust recorder functions and comprehensive logging of results for tracking setups and recordings.
- ❒ Full tracking of end suppression, defect marking capability, and customizable data retention.
- ❒ Seamless Integration with existing mill operations.



Echomac® electronics installed with an Echomac® UT 100 mm rotary for testing cold-finished carbon, alloy, and stainless bars.

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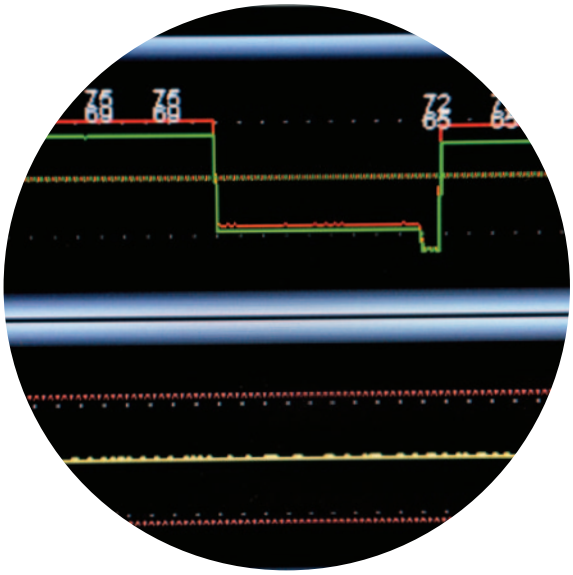
Echomac Operation and Control

Ultrasonic Control Panel - UT Screen

Provides full access and display of ALL ultrasonic test parameters. A-scan captures infrequent flaw echoes of short duration. Up to 4 measurement gates can be employed for each channel with graphic adjustment, live peak and other relevant test results are displayed real time. A Strip Chart provides an elapsed time linear display. All parameter settings from one channel can be easily copied to additional channels or adjust globally.

Multi Channel View - Multi Screen

Displays A-scan and Strip Chart of up to 32 individual channels or functional groups*, simultaneously. The strip chart shows the peak captured signal levels in color highlighted outlines, along with the numerical peak measurement within each gate. Graphic editing of visual devices such as gate, DAC, and scope position provide convenient adjustment.



Multi Screen Strip Chart view of
peak signal levels & numerical value



Tracking System - Track Screen

Accurately track product through Echomac® II test channels so each channel can be properly set for end suppression and flaw tracking through an encoder or simulated timer clock. Track screen provides control for all the parameters relating to the production line, alarm matrix routing, output control and sorting criteria. Complex arrangement and multiple line speed calibration are employed.

Production Recorder - Chart & Batch Screen

Strip charting and defect logging of all events for up to 32 individual channels or functional groups*, in both live or replay mode, is included as standard. Each chart clearly indicates Accept/Reject status. It also displays piece number and length, start time, date, line speed, and number of sample points taken. Batch screen manages record folders and production information input.



** A Functional group consists of channels with similar test functions such as channels for detecting longitudinal defects. These related channels are mapped into one chart for easy viewing, adjusting, or copying..*

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